

LESSON 2-8 • SESSION 2 • PRACTICE • FORM B

Reaching a Target Mean

Learning target – I can find the value a data set still needs in order to reach a target mean. • 6.SP.B.5c

REMEMBER

- **Target total = target mean $\times$ number of values.**
Count the missing value too.
- **Missing value = target total – sum so far.**
- If the missing value is more than the top score allowed, the goal is **not possible**.

WORKED EXAMPLE

Quiz scores are 6, 7 and 9. What fourth score gives a mean of 8?

Step 1 Count all scores, with the missing one: 4.

Step 2 Target total: $8 \times 4 = 32$. Sum so far: $6 + 7 + 9 = 22$.

Step 3 $32 - 22 = 10$. Check: $6 + 7 + 9 + 10 = 32$ and $32 \div 4 = 8$.

Answer The fourth score is 10.

WORDS target mean (*media objetivo*) • target total (*total objetivo*) • missing value (*valor faltante*)

PRACTICE

show your work • label units

1. Fill in each blank.

a. Mean 15, 6 values. Total = _____

b. Mean 9, 8 values. Total = _____

c. Need 84, have 66. Missing = _____

d. Need 130, have 102. Missing = _____

2. **Diving Contest.** A diver scored 26, 33, 28 and 31 points in four contests. He wants a mean of 30 points over five contests.

a. Target total: _____

b. Sum so far: _____

c. Contest five: _____

3. **Bowling Goal.** Rosa bowled 98, 106, 90 and 110. What score in game 5 gives a mean of 104?

Hint Turn the target mean into a target total first.

Show your work.

Score _____

4. **Missing Number.** Five numbers have a mean of 24. Four of them are 20, 27, 22 and 29. Find the fifth number.

Show your work.

Number _____

5. **One Perfect Quiz.** Marco's four quizzes have a mean of 75. He says a 100 on quiz 5 will make his mean 100. Find his real new mean.

Hint Rebuild the total, then share it across 5 quizzes.

Show your work.

New mean _____

6. **Working Backward.** Six temperatures have a mean of 52°F. Five of them are 48, 55, 50, 57 and 51°F. Find the sixth temperature.

Hint Multiply by 6, the count that includes the missing value.

Show your work.

Answer _____ °F

- 7. Impossible Target.** Lucía scored 72, 76 and 80 on three tests (top score 100). What score does she need for a mean of 92 over four tests? Is it possible?

Hint Compare the score needed with 100, the top score.

Show your work.

Score needed _____

- 8. Two Targets.** Three deliveries weighed 15, 22 and 17 kg. What must a fourth delivery weigh for each mean?

a. Mean of 20 kg: _____ kg

b. Mean of 22 kg: _____ kg

Show your work.

FIND THE ERROR

explain the mistake, then fix it

- 9. Talia’s Work.** Talia has a total of 112 points after four games. She wants a mean of 32 over five games, so she wrote $112 - 32 = 80$. Explain her mistake and find the score she needs.

TEST PRACTICE

choose the best answer

- 10.** Five values total 200. Their mean is –

A 1,000 **B** 40 **C** 195 **D** 50

- 11.** To reach a target mean, first find the –

A target total **B** median **C** greatest value **D** range

SELF-CHECK CODE

find your answer · copy its letter into that problem’s box · spell a math word

51 I	26 G	116 I	22 S	100 Q	228 X	104 Z	140 N	80 S	32 M
#2c		#3	#4	#5	#6	#7	#8a		

CHECK YOURSELF ☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____